Mendel’s labors

During the course of his research (1856–63), Mendel grew nearly 30,000 pea plants over several generations and carefully recorded the results. He focused on traits (phenotypes) that had only two distinct forms—for example, white or purple flowers. When examining the trait of yellow or green peas, Mendel took green pea plants and cross-pollinated them with yellow pea plants. The peas produced from this parent generation were all yellow and Mendel named them the F1 generation. He then cross-pollinated pea plants from the F1 generation with each other to produce the F2 generation. He found that some peas produced were yellow and some were green. The F1 generation showed only one trait (yellow), which Mendel called “dominant.” However, in the F2 generation 75 percent had the dominant yellow trait and 25 percent displayed the nondominant—or “recessive”—green trait.

“Heredity provides for the modification of its own machinery.”

James Mark Baldwin

American psychologist

Laws of inheritance

Mendel theorized that every pea plant has two factors controlling each trait. When plants are cross-pollinated, one factor is inherited from each plant. A factor can be dominant or recessive. When both inherited factors are dominant, the resulting plant will show the dominant trait. With a pair of recessive factors, the plant will show the recessive trait. However, if one dominant and one recessive factor are present, the plant will show the dominant trait.